

# ***Context-Aware Malay-to-English Translation with Malaysian Regional Dialect Support***

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# PROBLEM STATEMENT

Malaysia is a 14-state country where almost each and every state has its own unique dialect. However existing translation system struggles to understand regional dialects, informal slangs/expressions and has no sense of context.



# OBJECTIVES

## CONTENT

- Build a context-aware translation model
- Improve translation accuracy for dialects, slangs and have better understanding of context

## TARGET USERS



Tourist



Local Malaysian



Everyone

# RELATED WORKS

## NusaX

12 Apr 2023

### NusaX: Multilingual Parallel Sentiment Dataset for 10 Indonesian Local Languages

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#### Abstract

Natural language processing (NLP) has significant impact on society via technologies such as machine translation and search engines. Despite its success, NLP technology is only widely available for high-resource languages such as English and Mandarin Chinese, and remains inaccessible to many languages due to the unavailability of data resources and benchmarks. In this work, we focus on developing resources for languages of Indonesia. Despite being the second most linguistically-

The **NusaX** dataset was created by translating SmSA (Purwarianti and Crisdayanti, 2019) — an existing Indonesian sentiment analysis dataset containing comments and reviews from the IndoNLU benchmark (Wilie et al., 2020) — using competent bilingual speakers, coupled with additional human-assisted quality assurance. Sentiment analysis is one of the most popular NLP tasks, and has been explored in many applications in Indonesia, including presidential elections (Ibrahim et al., 2015; Budiharto and Meiliana, 2018), product reviews (Fauzi, 2019), stock forecasting (Cakra and

# RELATED WORKS

## NusaX

### ➡ WHAT THEY DID?

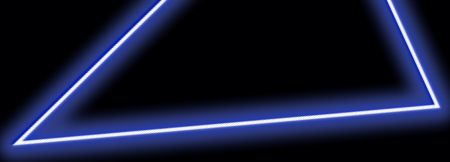
- Took an existing dataset of Indonesian sentences.
- Find native speakers in both Indonesian and local dialect.
- Make a dataset using bridge method (Dialect-Indonesian-English)

### ➡ WHAT IS THEIR RESULT?

- Developed a high-quality dataset NusaX which covers a lot of Indonesian dialect - English translation
- Explore which model that performs the best using their dataset.

### ➡ HOW CAN WE IMPLEMENT?

- Use the bridge method for our dataset.
- We did not just want to made the dataset, but also a model that compares to existing model such as mBERT and MalayBERT.
- Implement other dialects into our project.



# DATASET & EVAL



## DATA SOURCES

- standard Malay-English parallel data
- manually collected dialect/slang examples
- social media/ forums/ local phrase lists
- possible small custom dataset from native speakers

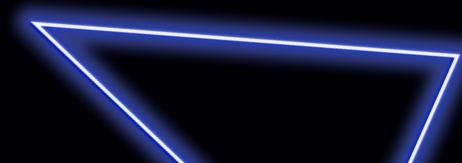
## DIALECT FOCUS

- Kelantan
- Terengganu
- Perak
- maybe others if time allows

## EVALUATION

- Automatic metrics : BLEU / chrF / BERTScore / COMET
- Human evaluation : (correctness/ naturalness/ context appropriate)

## BASELINES

- MalayBERT
  - mBERT
  - Google Translate
  - model without dialect / context support
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# PROJECT OUTLINE

## ⇒ PIPELINE

1. Input Malay sentence (Query)
2. Detect whether it contains dialect/slang
3. Normalize or tag dialect words
4. Pretrained model using bridge method will translate into English
5. Compare with baseline translation

# EXPECTED OUTCOME

1

- Better translation of dialect/slang than baseline

2

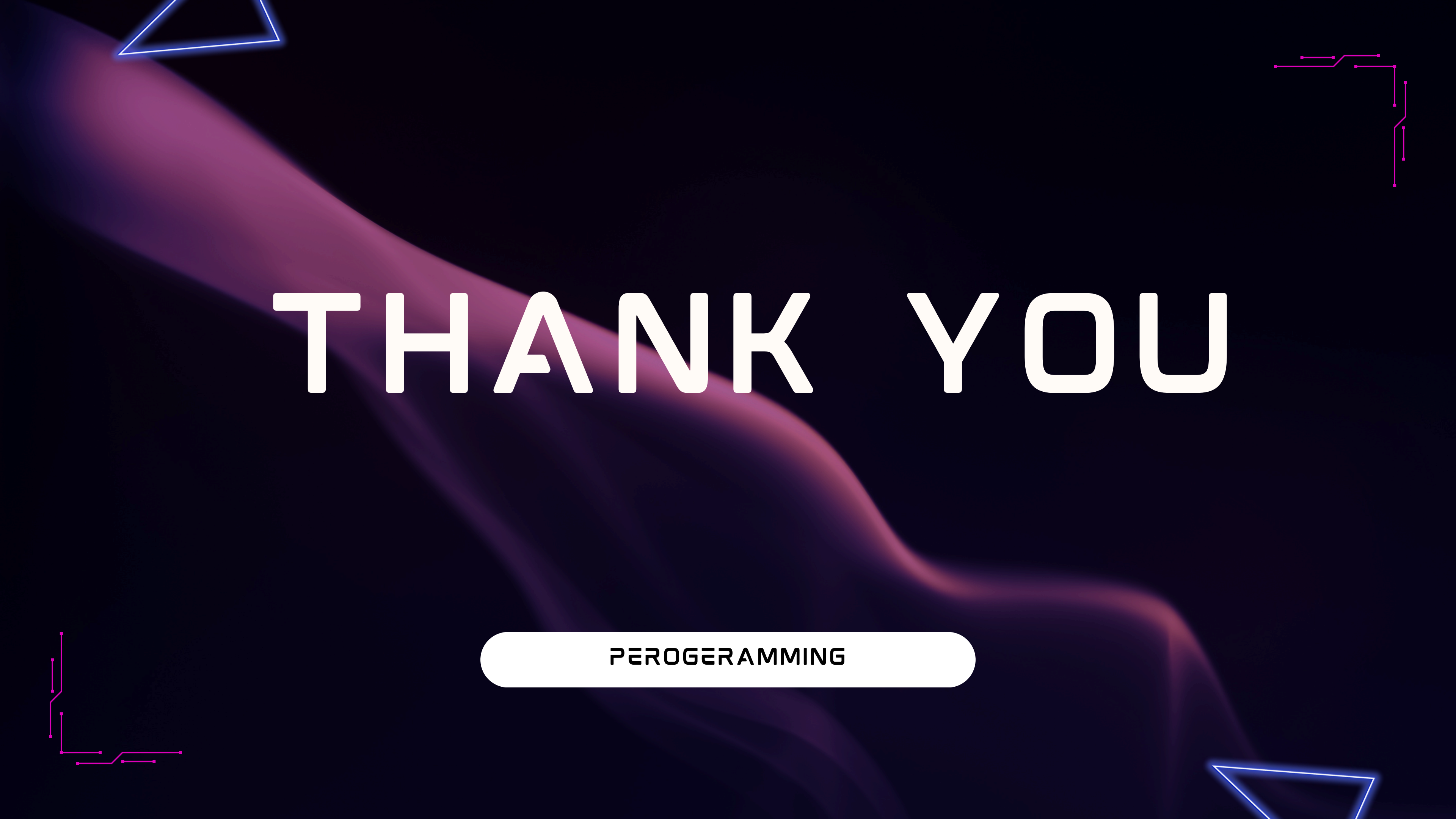
- fewer wrong literal translation and more natural translation

3

- develop a dataset for Dialect-Malay-English

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THANK YOU

PEROGGERAMMING